Eigenvalues and Eigenvectors

Math Foundations — Node 07

1 Motivation

A linear map $A : \mathbb{R}^n \to \mathbb{R}^n$ is, in general, a complicated object: it mixes coordinates, rotates, shears, and rescales. The whole point of eigentheory is to look for the directions on which A acts trivially — not literally as the identity, but as pure scaling. Once such a direction v has been found, the action of A along it is described by a single number λ, the corresponding eigenvalue, via $Av = \lambda v$. If enough such directions can be found to form a basis, then A itself factorises as a change of basis followed by a diagonal scaling. This is the cleanest possible structural description of a linear map and it underlies essentially every spectral technique in numerical analysis, optimisation, and machine learning.

2 Definitions

Throughout, let $\mathbb{F}$ denote either $\mathbb{R}$ or $\mathbb{C}$ and let $A \in \mathbb{F}^{n\times n}$.

Definition 1 (Eigenvalue, eigenvector)**.** A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of A if there exists a non-zero vector $v \in \mathbb{F}^n$ such that

$$Av = \lambda v.$$

Any such v is called an *eigenvector* of A associated with the eigenvalue λ. The set of all eigenvectors for a fixed λ together with the zero vector is a linear subspace of $\mathbb{F}^n$ called the *eigenspace* $E_\lambda = \ker(A - \lambda I)$. The set of all eigenvalues of A is the *spectrum* $\sigma(A)$.

Remark. The condition $v \neq 0$ is essential. Without it, every scalar λ would satisfy $A \cdot 0 = \lambda \cdot 0$ trivially, and the definition would carry no information.

Definition 2 (Characteristic polynomial)**.** The *characteristic polynomial* of A is

$$p_A(\lambda) = \det(A - \lambda I) \in \mathbb{F}[\lambda].$$

It is a polynomial of degree exactly n in λ with leading term $(-1)^n\lambda^n$.

Definition 3 (Algebraic and geometric multiplicity)**.** Let $\lambda_0 \in \sigma(A)$.

- The *algebraic multiplicity* $m_a(\lambda_0)$ is the multiplicity of λ_0 as a root of p_A.
- The *geometric multiplicity* $m_g(\lambda_0)$ is the dimension of the eigenspace $E_{\lambda_0} = \ker(A - \lambda_0 I)$.

It is always the case that $1 \le m_g(\lambda_0) \le m_a(\lambda_0)$.

Definition 4 (Diagonalisability)**.** A is *diagonalisable* over $\mathbb{F}$ if there exist an invertible $V \in \mathbb{F}^{n\times n}$ and a diagonal $\Lambda \in \mathbb{F}^{n\times n}$ such that $A = V\Lambda V^{-1}$. Equivalently, $\mathbb{F}^n$ admits a basis of eigenvectors of A, equivalently $m_a(\lambda) = m_g(\lambda)$ for every $\lambda \in \sigma(A)$.

3 The fundamental theorem

We now state and prove the main structural fact tying eigenvalues to roots of the characteristic polynomial.

Theorem 5 (Eigenvalues are roots of the characteristic polynomial)**.** *Let $A \in \mathbb{F}^{n\times n}$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A if and only if*

$$\det(A - \lambda I) \;=\; 0.$$

Proof. We prove both implications.

($\Rightarrow$). Suppose λ is an eigenvalue of A. By Definition 1 there exists a non-zero vector $v \in \mathbb{F}^n$ with $Av = \lambda v$. Rearranging,

$$(A - \lambda I)v \;=\; Av - \lambda v \;=\; 0.$$

Thus the homogeneous linear system $(A - \lambda I)x = 0$ has the non-trivial solution $x = v \neq 0$. A square matrix admits a non-trivial kernel vector if and only if it is singular, equivalently iff its determinant vanishes. Therefore $\det(A - \lambda I) = 0$.

($\Leftarrow$). Conversely, suppose $\det(A - \lambda I) = 0$. Then $A - \lambda I$ is a singular $n \times n$ matrix over $\mathbb{F}$, so its kernel $\ker(A - \lambda I)$ is a non-trivial subspace of $\mathbb{F}^n$ (it has positive dimension, by the rank–nullity theorem applied to the linear map $x \mapsto (A - \lambda I)x$, whose rank is at most $n - 1$). Choose any non-zero $v \in \ker(A - \lambda I)$. Then $(A - \lambda I)v = 0$, i.e. $Av = \lambda v$, with $v \neq 0$. By definition, λ is an eigenvalue of A with eigenvector v. □

Corollary 6. *A matrix $A \in \mathbb{C}^{n\times n}$ has at least one eigenvalue in $\mathbb{C}$. In fact it has exactly n eigenvalues counted with algebraic multiplicity, because $p_A \in \mathbb{C}[\lambda]$ splits completely over $\mathbb{C}$ by the fundamental theorem of algebra.*

4 The spectral theorem (symmetric case)

Theorem 7 (Real spectral theorem)**.** *If $A \in \mathbb{R}^{n\times n}$ is symmetric ($A^\top = A$), then all eigenvalues of A are real, and there exists an orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ (i.e. $Q^\top Q = I$) and a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ such that*

$$A \;=\; Q\Lambda Q^\top.$$

We do not prove this here but record it because it is the workhorse fact for covariance matrices, Gram matrices, and Laplacians.

5 A small worked example

Example 8. Let

$$A \;=\; \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$p_A(\lambda) \;=\; \det\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} \;=\; (2-\lambda)^2 - 1 \;=\; \lambda^2 - 4\lambda + 3 \;=\; (\lambda - 1)(\lambda - 3).$$

Thus $\sigma(A) = \{1, 3\}$. Solving $(A - I)v = 0$ gives $v_1 = (1, -1)^\top/\sqrt{2}$, while $(A - 3I)v = 0$ gives $v_2 = (1, 1)^\top/\sqrt{2}$. These are orthonormal, as guaranteed by Theorem 7, and

$$A \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

6 Where to go next

The next step in this map is the singular value decomposition (SVD), which extends eigendecomposition to rectangular and non-normal matrices and underlies LoRA-style low-rank adaptation, PCA, and dimension reduction in general.